SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

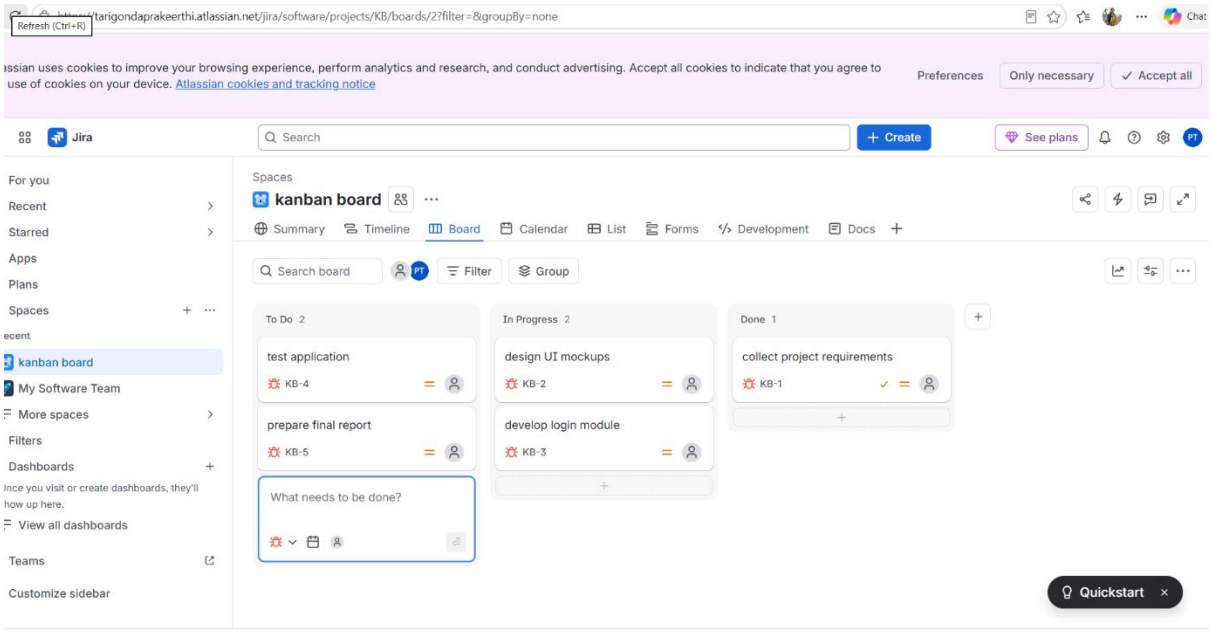
CSA1012 - Software Engineering **Name: T.Prakeerthi**

Jira Lab Experiment **Reg.No:192524279**

Experiment 1

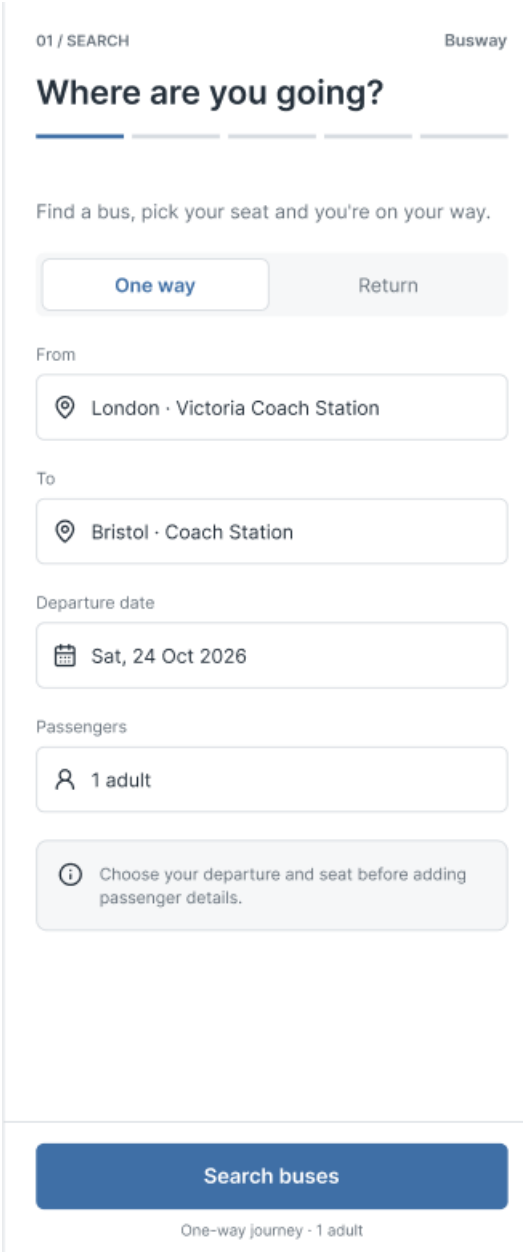
1.Create a Kanban Board to Visualize the Tasks.

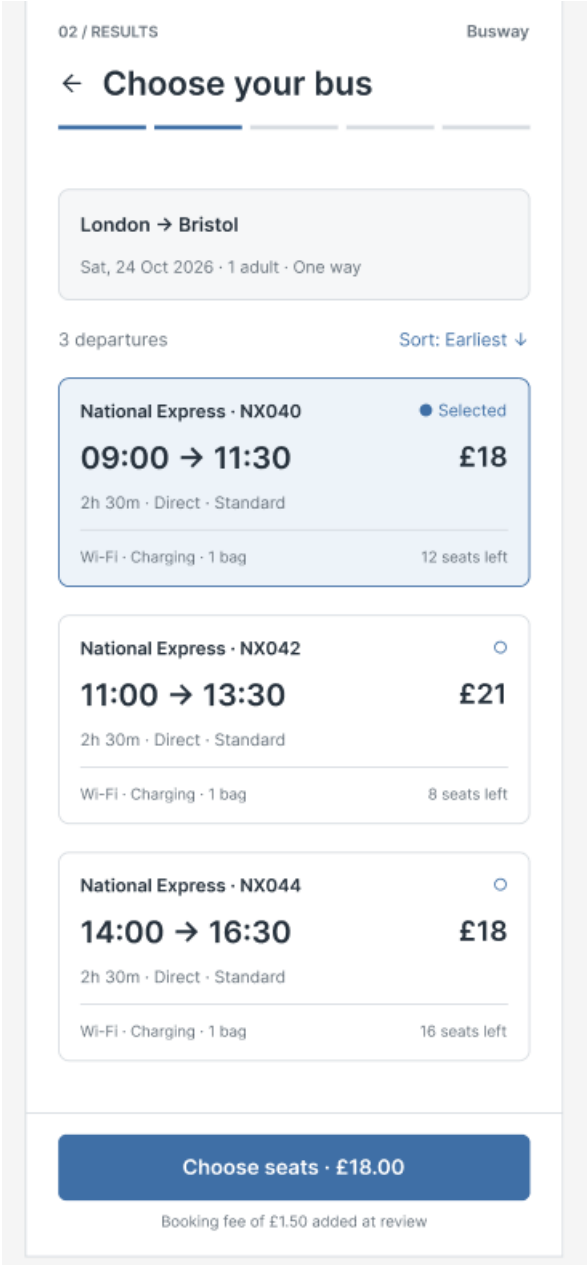
- **Create Columns for To Do, In-Progress and Done.**
- **Add Atleast 5 Sample Tasks**
- **Move the Tasks across the Columns to Simulate the Workflow.**

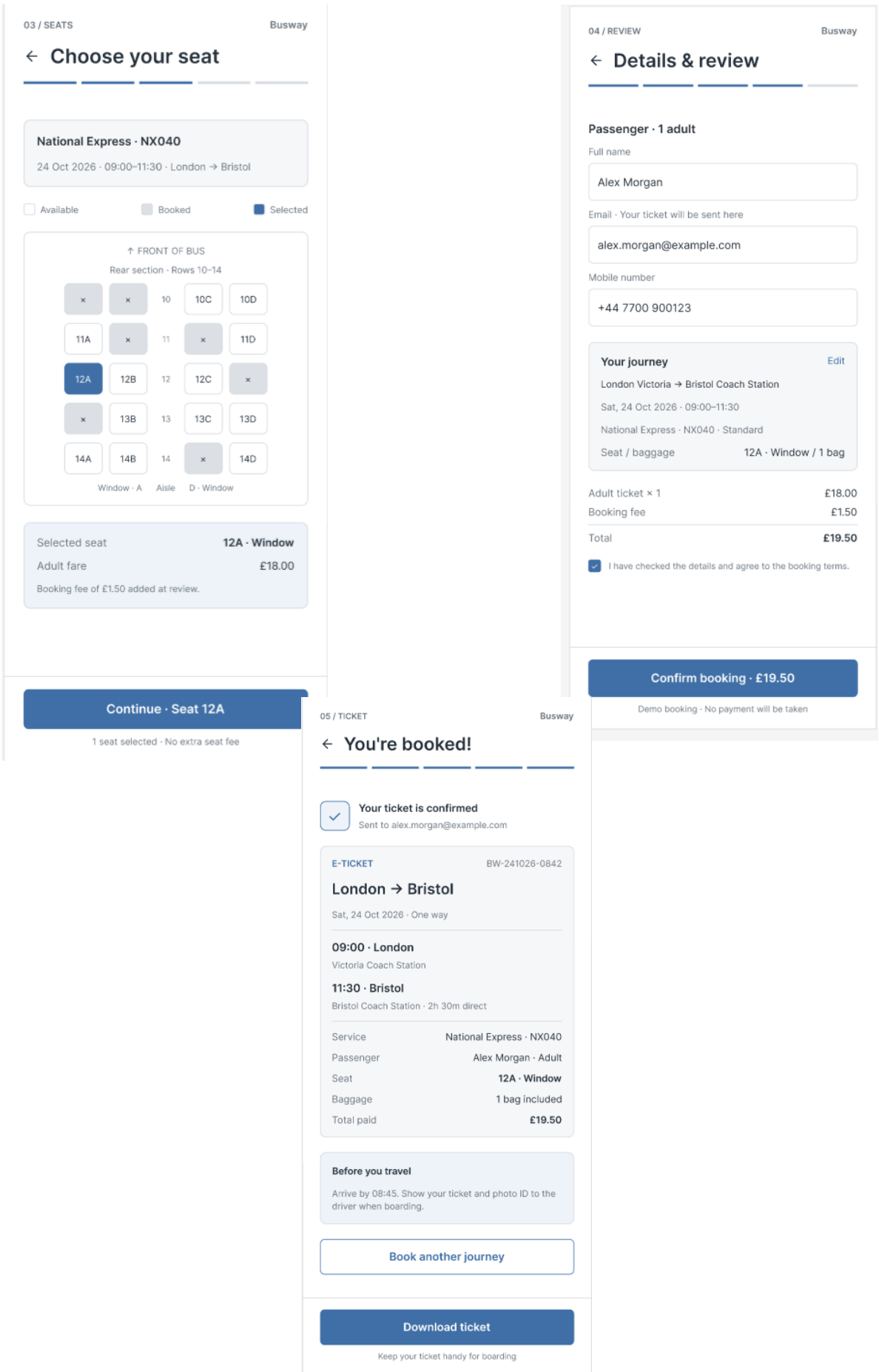


Experiment 2

2.Sketch a Simple Prototype of a Bus Ticket Booking System using Figma Tool

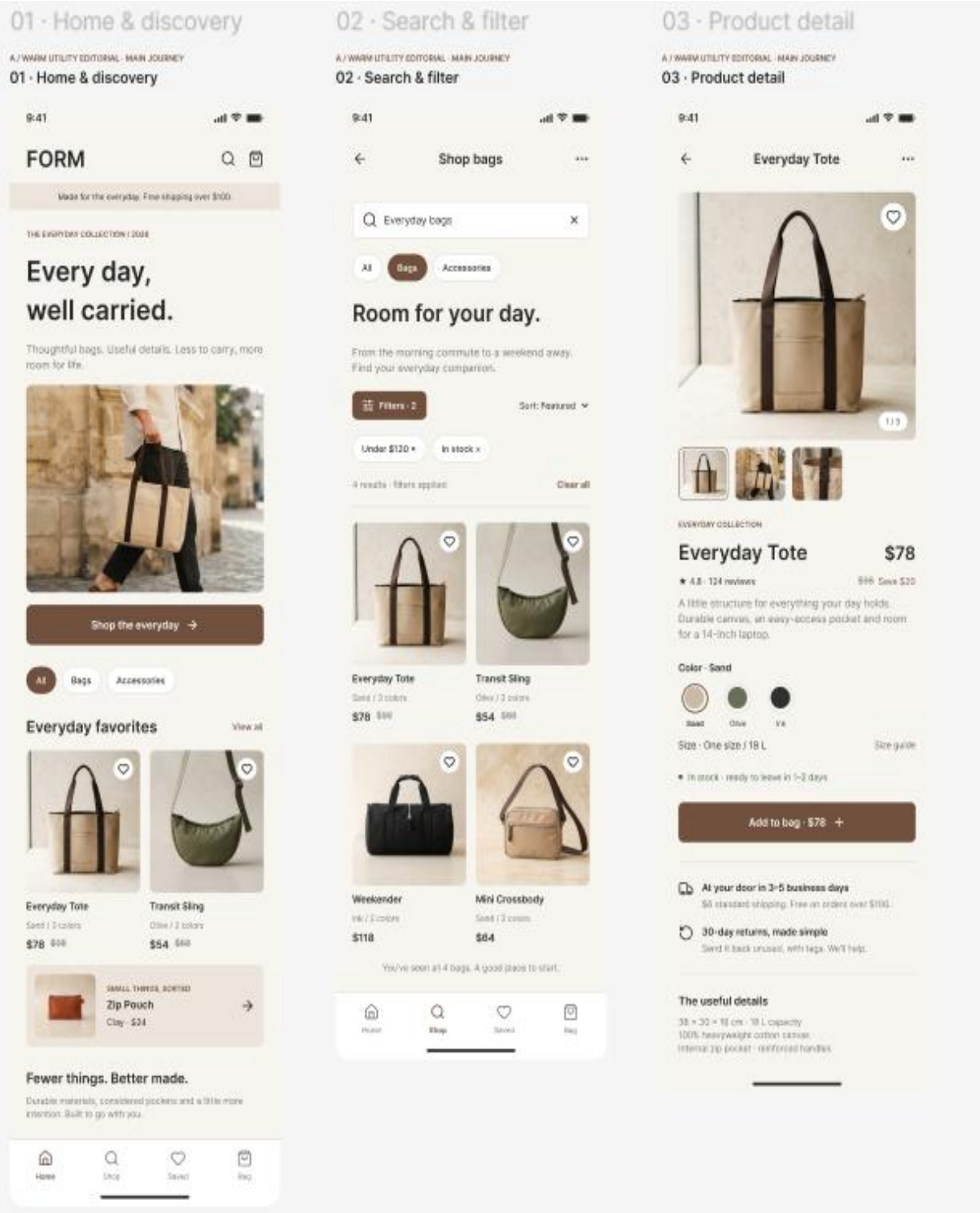


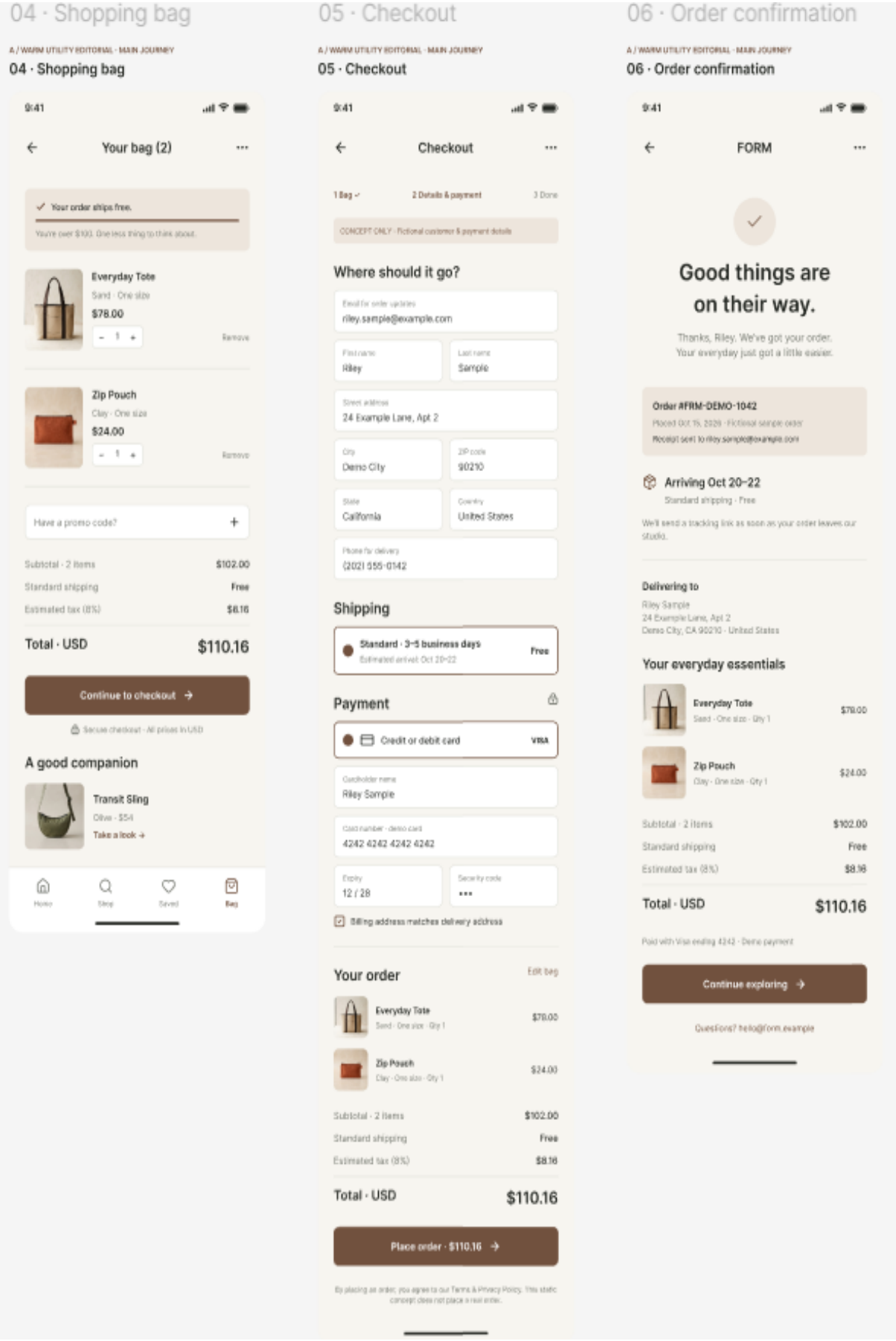



03 / SEATS
Busway
Choose your seat
National Express · NX040
24 Oct 2026 · 09:00–11:30 · London → Bristol
Available
Booked
Selected
↑ FRONT OF BUS
Rear section · Rows 10–14
10C
10D
11A
11D
12A
12B
12C
13B
13C
13D
14A
14B
14D
Window · A
Aisle
D · Window
Selected seat
12A · Window
Adult fare
£18.00
Booking fee of £1.50 added at review.
Continue · Seat 12A
1 seat selected · No extra seat fee
04 / REVIEW
Busway
Details & review
Passenger · 1 adult
Full name
Alex Morgan
Email · Your ticket will be sent here
alex.morgan@example.com
Mobile number
+44 7700 900123
Your journey
Edit
London Victoria → Bristol Coach Station
Sat, 24 Oct 2026 · 09:00–11:30
National Express · NX040 · Standard
Seat / baggage
12A · Window / 1 bag
Adult ticket × 1
£18.00
Booking fee
£1.50
Total
£19.50
I have checked the details and agree to the booking terms.
Confirm booking · £19.50
Demo booking · No payment will be taken
05 / TICKET
Busway
You're booked!
Your ticket is confirmed
Sent to alex.morgan@example.com
E-TICKET
BW-241026-0842
London → Bristol
Sat, 24 Oct 2026 · One way
09:00 · London
Victoria Coach Station
11:30 · Bristol
Bristol Coach Station · 2h 30m direct
Service
National Express · NX040
Passenger
Alex Morgan · Adult
Seat
12A · Window
Baggage
1 bag included
Total paid
£19.50
Before you travel
Arrive by 08:45. Show your ticket and photo ID to the driver when boarding.
Book another journey
Download ticket
Keep your ticket handy for boarding

Experiment 3

3.The stakeholders have conflicting views on the user interface design for an E-Commerce mobile app. Create a prototype using Figma tool to discuss with the stakeholders to get their feedback and approval.

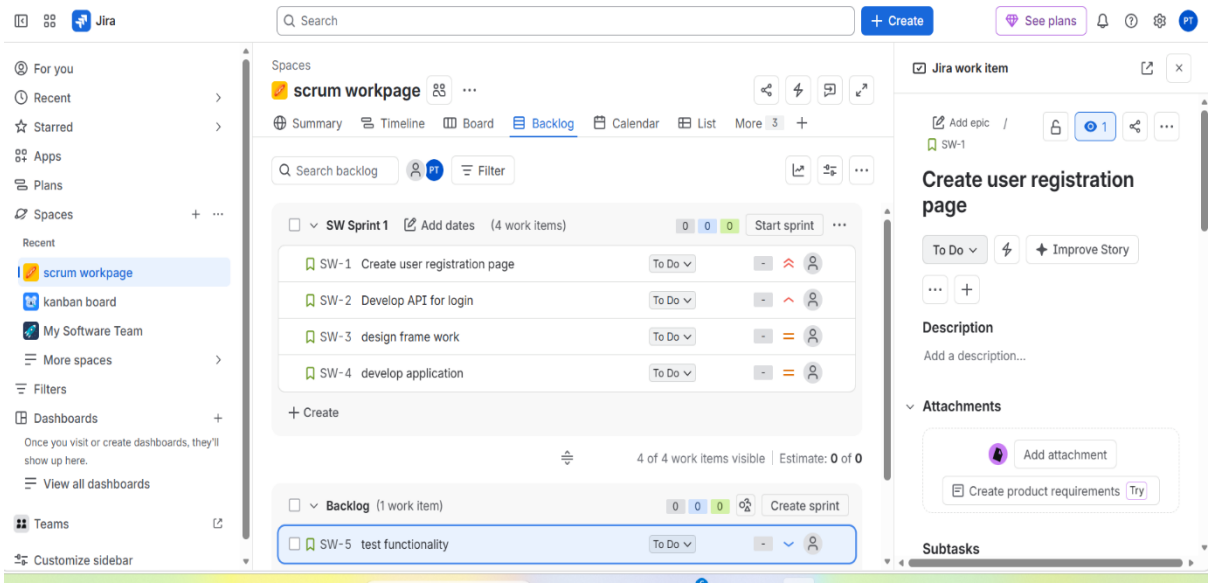


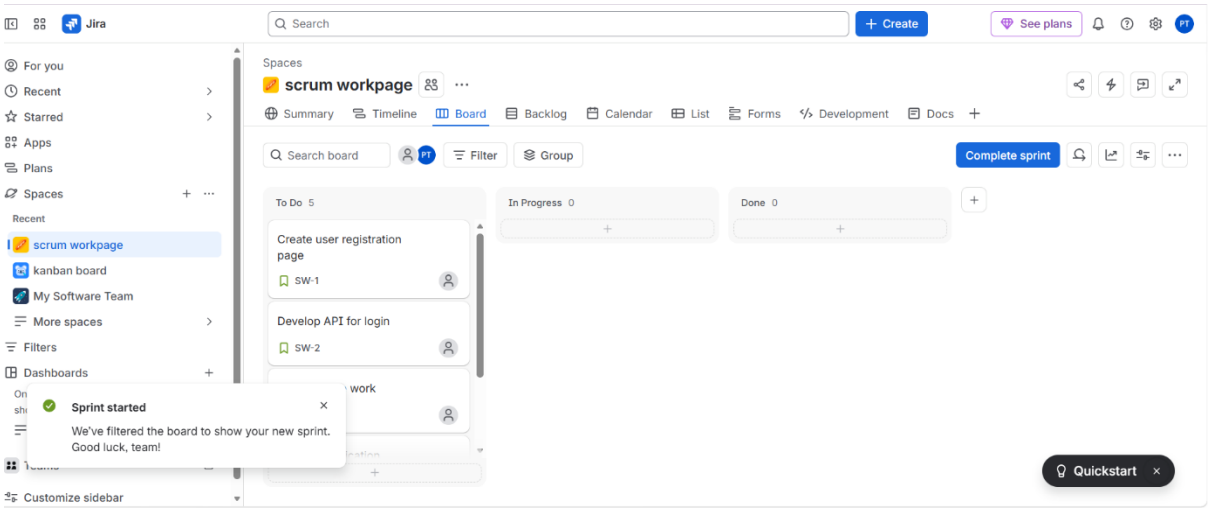

04 · Shopping bag
A / WARM UTILITY EDITORIAL · MAIN JOURNEY
04 · Shopping bag
9:41
Your bag (2)
Your order ships free.
Everyday Tote
Sand · One size
$78.00
Remove
Zip Pouch
Clay · One size
$24.00
Remove
Have a promo code?
Subtotal · 2 items
$102.00
Standard shipping
Free
Estimated tax (8%)
$8.16
Total · USD
$110.16
Continue to checkout
Secure checkout · All prices in USD
A good companion
Transit Sling
Olive · $54
Take a look
Home
Shop
Saved
Bag
05 · Checkout
A / WARM UTILITY EDITORIAL · MAIN JOURNEY
05 · Checkout
9:41
Checkout
1 Bag
2 Details & payment
3 Done
CONCEPT ONLY · Fictional customer & payment details
Where should it go?
riley.sample@example.com
Riley
Sample
24 Example Lane, Apt 2
Demo City
90210
California
United States
(202) 555-0142
Shipping
Standard · 3–5 business days
Free
Payment
Credit or debit card
VISA
Riley Sample
4242 4242 4242 4242
12 / 28
Billing address matches delivery address
Your order
Edit bag
Everyday Tote
$78.00
Zip Pouch
$24.00
Subtotal · 2 items
$102.00
Standard shipping
Free
Estimated tax (8%)
$8.16
Total · USD
$110.16
Place order · $110.16
06 · Order confirmation
A / WARM UTILITY EDITORIAL · MAIN JOURNEY
06 · Order confirmation
9:41
FORM
Good things are on their way.
Thanks, Riley. We've got your order.
Your everyday just got a little easier.
Order #FRM-DEMO-1042
Arriving Oct 20–22
Standard shipping · Free
Delivering to
Riley Sample
24 Example Lane, Apt 2
Demo City, CA 90210 · United States
Your everyday essentials
Everyday Tote
$78.00
Zip Pouch
$24.00
Subtotal · 2 items
$102.00
Standard shipping
Free
Estimated tax (8%)
$8.16
Total · USD
$110.16
Continue exploring
Questions? hello@form.example

Experiment 4

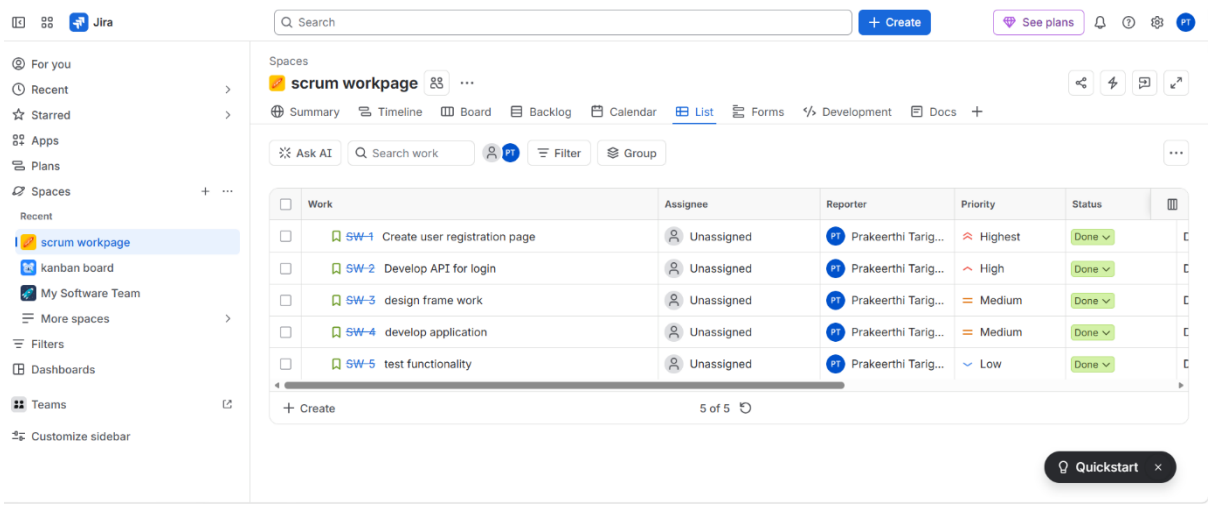
4.Create a Scrum Project in Jira.

- Add a backlog with at least 5 items (e.g., "Create user
- registration page", "Develop API for login").
- Prioritize the backlog and create a 1-week sprint.
- Move backlog items into the sprint and start the sprint.
- Finally show the Screenshot





SOFTWARE ENGINEERING LAB EXPERIMENTS



Experiment 5

5.Use the following requirements for a Library Management System:

- Add a feature to search books by title and author.
- Implement an online book reservation system.
- Generate monthly reports on borrowed books for
- administrators.
- Enable email notifications for overdue books.
- Add support for QR code scanning for borrowing and
- returning books.
- Create a user-friendly dashboard for librarians.
- Allow users to review and rate books.
- Integrate a chatbot for user assistance.
- Develop a mobile app version of the system.
- Provide multi-language support.
- Categorize each requirement using MOSCOW Method (Must-Have,
- Should-Have, Could-Have, or Won't-Have) based on the following
- criteria:
- Impact on the users and stakeholders.
- Feasibility considering time, budget, and resource constraints.
- Finally Submit the completed Google Sheet or Excel file with all
- requirements categorized and justified.

A	B	C	D	E	F
No	Requirement	MoSCoW Category	Impact on Users/Stakeholders	Feasibility	Justification
1	Search books by title and author	Must-Have	High — users need to find books quickly; librarians handle fewer basic search requests.	High — a basic search can be built using the book catalogue database and a simple search interface.	This is a core library function and is relatively inexpensive to implement, so it should be included in the first release.
2	Online book reservation system	Must-Have	High — users can reserve unavailable or in-demand books without visiting the library.	Medium to High — requires reservation rules, availability checks, and a database; achievable in the main project.	It directly supports the main purpose of the system and provides important value to users.
3	Monthly reports on borrowed books for administrators	Should-Have	High for administrators — helps monitor borrowing trends and make decisions.	High — reports can be generated from borrowing transaction records using date filters and summaries.	Useful for management, but the system can begin operating without automated monthly reports; basic transaction records should still be stored.
4	Email notifications for overdue books	Should-Have	Medium to High — reminds users to return books and helps reduce overdue loans.	Medium — requires email service configuration, notification templates, and scheduled checks.	Valuable for improving returns, but manual reminders can be used temporarily if email integration is delayed.
5	QR code scanning for borrowing and returning books	Could-Have	Medium — speeds up checkout and return operations and reduces manual entry.	Medium — requires QR code generation, a scanner or camera interface, and transaction integration.	It improves convenience but is not essential because librarians can process loans and returns using book IDs or barcodes.
6	User-friendly dashboard for librarians	Must-Have	High — librarians need a clear way to manage books, users, loans, returns, and reservations.	High — a basic dashboard can be developed with standard web interface components.	A simple operational dashboard is necessary for staff to use and manage the system effectively.

N	Requirement	MoSCoW Category	Impact on Users/Stakeholders	Feasibility	Justification
6	User-friendly dashboard for librarians	Must-Have	High — librarians need a clear way to manage books, users, loans, returns, and reservations.	High — a basic dashboard can be developed with standard web interface components.	A simple operational dashboard is necessary for staff to use and manage the system effectively.
7	Allow users to review and rate books	Could-Have	Low to Medium — reviews help readers choose books and provide feedback.	Medium — requires review storage, rating calculations, and moderation or abuse controls.	This is an extra engagement feature and can be added after essential library operations are stable.
8	Chatbot for user assistance	Won't-Have (for this release)	Medium — could answer common questions, but users can initially use help pages or contact library staff.	Low to Medium — a reliable chatbot may need integration, content preparation, testing, and ongoing maintenance.	Defer from the current release to control cost and complexity. This does not mean the feature can never be developed.
9	Mobile app version of the system	Won't-Have (for this release)	High for mobile users — provides convenient access on phones.	Low to Medium under limited resources — a separate app adds design, development, testing, and maintenance work.	Prioritize a responsive website first so mobile users can still access core functions; consider a dedicated app in a later release.
10	Multi-language support	Could-Have	Medium to High in a multilingual community — makes the system more accessible.	Medium — requires translated interface text, language selection, and testing in each supported language.	It is beneficial, but can follow the first release if time and translation resources are limited.

Git Hub Link:

https://github.com/tarigondaprakeerthi/CSA1012-SOFTWARE-ENGINEERING-.git

SOFTWARE ENGINEERING LAB EXPERIMENTS

Lab Experiments